

Final Report – Airbnb Rental Prices

Our goal with this project was to find a model that does the best job of predicting the price of each Airbnb rental based on variables like number of bathrooms or bedrooms, how many people can be accommodated, and more. With our best model, the most important predictors are number of bathrooms, how many people can be accommodated, number of bedrooms, number of host listings, and number of reviews. This best model resulted in predictions that were, on average, about \$81 off from the true prices, compared to the other models we fitted generally being about \$160-\$170 off from the true prices on average.

However, we noticed a lot of rental postings – over 8% of all measured – that had unusually high prices compared to other postings with similar statistics. This makes sense because prices can tend to have some unusually high values – or outliers. Because of these outliers the predictions on the data can sometimes be off. Additionally, some of the predictors were quite closely related; for example, as the number of people that could be accommodated increased, so did the number of bedrooms. While this makes sense, we wanted to offset the fact that some predictors explained the same information, so the model we chose was intended to circumvent this correlation.

In the future, to try to explain the presence of so many outliers, it could be beneficial to collect data on more different variables to see if any of them could explain the higher prices.

Key Takeaways

- 1) Some of the important predictors of rental price are number of bathrooms, how many people can be accommodated, number of bedrooms, number of host listings, and number of reviews.

- 2) On average, our best model predicts prices to be around \$81 off from the actual price of each rental.
- 3) The high number of outliers in the dataset could result in predictions being off, so our model was created taking this into account.